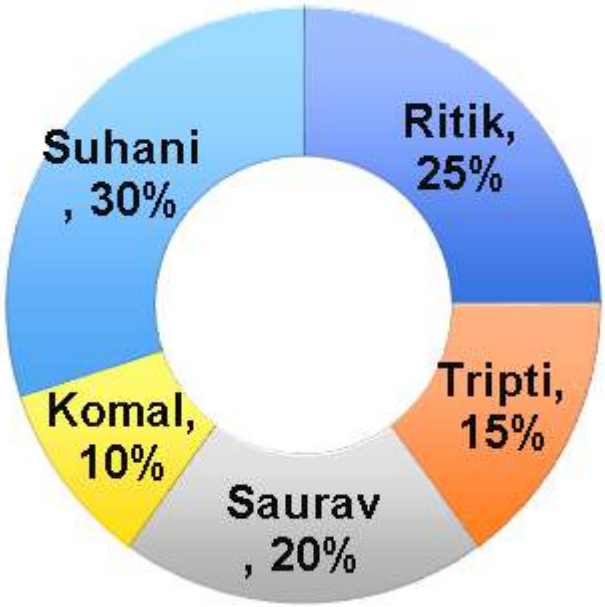
Data Interpretation for Banking Exams

8. Doughnut shows the percentage of the ages of five different persons.
   Total= 120



i) The average ages of Tripti, Anju and Afreen is 18 years and Suhani's age is 50% more than Afreen's. The average ages of Anju and Suja after 6 years is 25 years. What is Suja's age 10 years ago?

a) 12 yrs b) 14 yrs c) 26 yrs d) 16 yrs e) 24 yrs

ii) The present age of Anju is 50% more than the present age of Afreen. If the average age of Afreen, Komal and Suja is 24 years and the ratio of the ages of Suja to Ritik is 7:5, then what is the average age of Anju, Afreen and Suja?

a) 29 yrs b) 27 yrs c) 20 yrs d) 25 yrs e) 21 yrs

iii) Average ages of Saurav, Ritik and Suhani is equal to the sum of the ages of Suruchi and Sonal. If the product of the ages of Suruchi and Komal is 156 yrs, then what is the present age of Sonal?

a) 23 yrs b) 14 yrs c) 15 yrs d) 17 yrs e) 18 yrs

iv) The difference between the ages of Komal and Saurav is equal to present age of Anju. Ratio of the ages of Anju and Afreen after 8 yrs is 2:3. Find Afreen's age 9 yrs ago?

a) 12 yrs b) 13 yrs c) 14 yrs d) 16 yrs e) 17 yrs

v) The difference between the ages of Ritik and Komal is equal to Anju's age 7 yrs ago. If the ratio of the ages of Anju to Afreen after 5 years is 5:8 and Suja's present age is half of Afreen's age one year ago. What is the difference between the average ages of Suja, Komal and Suhani together and the average ages of Saurav and Afreen together?

a) 11.5 yrs b) 15.5 yrs c) 10.5 yrs d) 16.4 yrs e) 17.5 yrs

20th March 2026

Q8

	%	Age
Ritik	25%	30 yrs
Tripti	15%	18 yrs
Saurav	20%	24 yrs
Komal	10%	12 yrs
Suhani	30%	36 yrs
Total	100%	

$100\% = 120$

$1\% = \frac{120}{100}$

(i) Avg age of Tripti, Anju & Afreen = 18 yrs

$\therefore$ Sum = $18 \times 3 = 54$ yrs

Tripti's age = 15% = $15 \times \frac{120}{100}$ = 18 yrs (present)

$\underbrace{\text{Tripti} + \text{Anju} + \text{Afreen}}_{} = 54$

18 ↓ 12 36 ↓ 24

Suhani = $\frac{3}{2} \times$ Afreen

$36 \times \frac{2}{3}$ = Afreen

Suhani = 30% = $30 \times \frac{120}{100}$

= 36 yrs.

$\therefore$ Afreen = 24 yrs. (Present)

$\therefore$ Anju = (36 − 24) = 12 yrs. (Present)

Present (Anju + Suja) 38 (Present)

$\downarrow$ +6 yrs $\qquad$ $\uparrow$ −6 yrs.

After 6 yrs $\qquad$ 50 $\qquad$ Avg = 25 (Anju + Suja)

$\rightarrow \therefore$ Sum = 25 × 2 = 50

Anju + Suja = 38

12 + Suja = 38

Suja = 38 − 12

Suja = 26 yrs. (Present)

$\therefore$ Suja's age 10 years ago = 26 − 10 = 16 Ans.

(iii) Avg → $\frac{\text{Saurav + Ritik + Suhani}}{3}$ = Suruchi + Sonal

$\frac{24 + 30 + 36}{3}$ = 13 + Sonal

$\frac{\cancel{90}\ 30}{\cancel{3}}$ = 13 + Sonal

30 = 13 + Sonal

Suruchi × Komal = 156

Suruchi × $\cancel{12}$ = $\cancel{156}$ 13

Suruchi = 13

$\therefore$ Sonal = 30 − 13 = 17 yrs. Ans.

Quadratic Equations for Banking Exams

QUADRATIC EQUATIONS

Relations b/w x and y:

* To get the relation of x and y, we need the roots of x and y.

BASIC WAY:

Eg: $5x^2 + 15$

$= 5(x^2+3)$

5, x^2+3 > factors of $5x^2+15$

To find factors by SPLITTING THE MIDDLE TERM:

Eg: (i) $x^2 + 5x + 6$

(split) $\longrightarrow$ sum $= +5x$

$x^2 + 2x + 3x + 6$

product $= +6 \times x^2$

$= 6x^2$

1×6

$x(x+2) + 3(x+2)$

factors

$(x+2), (x+3) = 0$

Sum $= 5x \rightarrow 6x - 1x = 5x$

Product $= 6x^2 \rightarrow 6x \times -1x = -6$

$x+2=0$ | $x+3=0$

$\boxed{x=-2}$ | $\boxed{x=-3}$

2×3

$2x + 3x = 5x$

$2x \times 3x = 6x^2$

RELATIONS:

(i) $x > y$

(ii) $x < y$

(iii) $x \geq y$

(iv) $x \leq y$

(v) $x = y$ or R.C.E (Relationship cannot be established)

Date :

TYPES OF SINGS (what we can get in an equations.

EQUATIONS		ROOTS	
−	+	+	+
+	+	−	−
+	−	−	+
−	−	+	−

} GOLDEN TRICK (RCE)

Eg: ① $x^2 + 5x + 6$ — Equation signs

$x = -2$, $x = -3$ — Root signs.

$x^2 - 20x + 96 = 0$

Roots = + + (x_1, x_2)

$y^2 + 32y + 231 = 0$

Roots = − − (y_1, y_2)

Check relation :-

$x_1 > y_1$ $\quad$ $x_2 > y_1$

$x_1 > y_2$ $\quad$ $x_2 > y_2$

$$x > y$$

To find the Roots :-

$x^2 - 20x + 96 = 0$ → we'll have to add up two values that will give us 20x, and their products will give us 96.

write the greater no. 1st

+12 +8

96 → 1 × 96 ✗
2 × 48 ✗
3 × 32 ✗
4 × 24 ✗ → don't take −ve 20 (we don't (−))
6 × 16 ✗
8 × 12 ✓ → add up to get 20. → prod = 96.

$y^2 + 32y + 231 = 0$ → we have to add factors of 231 :

−11 −21

3	231
7	77
11	11
	1

21

$x > y$

*Q8. $x^2 = 49$ $\qquad$ $y^3 = 343$

$x > y$

$x < y$

$x \geq y$

$x \leq y$

$x = y$ or RCE

* when the power is even (x^2, x^4, x^6) you always get 2 roots (+ve & -ve)

* when the power is odd (x^3, x^5, x^7, etc) you will always get 1 root (+ve or -ve depends on the Ques)

$x = 7 \times 7 = 49$ $\qquad$ $y = 7 \times 7 \times 7 = 343$

$x = -7 \times -7 = 49$ $\qquad$ $y = -7 \times -7 \times -7 = -343$

$\therefore x = +7$ $\qquad$ $x = -7$ $\qquad$ $y = +7$

$<$

$=$

7th Apr. 2026

$x \leq y$ Ans

Based on SQ ROOTS

Q9. $x^2 - 15\sqrt{3}x + 162 = 0$; $y^2 - 9\sqrt{2}y + 36 = 0$

(+) (+) $\qquad$ (+) (+)

$15\sqrt{3}x + 162$ $\qquad$ $9\sqrt{2}y + 36$

SQUARE $\sqrt{3}$ $\qquad$ SQUARE $\sqrt{2}$

3 $\qquad$ 2

54 $\qquad$ 18

$+\cancel{9}\sqrt{3}$ $+\cancel{6}\sqrt{3}$ $\qquad$ $+\cancel{6}\sqrt{2}$ $+\cancel{3}\sqrt{2}$

3 $\qquad$ 2 $\qquad$ 2 $\qquad$ 1

① break down into two values such that when we add them, they give us 15 or 9

② (put $\sqrt{3}$ after) $\qquad$ (Put $\sqrt{2}$ after)

Not mandatory

③ (can cancel with 3 for all) → cancel whenever possible w/ common factor to ease calculations.

$+(3\sqrt{3})^2 + (2\sqrt{3})^2 \qquad +(2\sqrt{2})^2 + (\sqrt{2})^2$ → ④ Square all the values.

$+9 \times 3 \quad + 4 \times 3 \qquad + 4 \times 2 \quad + 2$

$+27 \quad + 12 \qquad + 8 \quad + 2$

$\therefore x > y$ Ans.

Percentage

Date: 23rd Feb 2026

PERCENTAGE

↳ Basis of most chapters.

PERCENT

per 100

% INTO FRACTION (÷100)

$20\% = \frac{20}{100} = \frac{1}{5}$

$30\% = \frac{30}{100} = \frac{3}{10}$

$25\% = \frac{25}{100} = \frac{1}{4}$

$16\frac{2}{3}\% = \frac{50}{3 \times 100} = \frac{1}{6}$

$11\frac{1}{9}\% = \frac{100}{9 \times 100} = \frac{1}{9}$

$6\frac{1}{4}\% = \frac{25}{4 \times 100} = \frac{1}{16}$

FRACTION INTO % (×100)

$\frac{1}{4} = \frac{1}{4} \times 100 = 25\%$

$\frac{1}{5} = \frac{1}{5} \times 100 = 20\%$

47. A's income is 25% more than income of B. Income of B is how much percent less than income of A?

(a) 10% (b) 20% (c) 40% (d) $16\frac{2}{3}\%$

QUESTIONS BASED ON AB RULE CONCEPT.

Q47. A's income = 25% more than B.

$25\% = +\frac{1}{4}$ (A more than B)

* Numerator has to be the same.

AB Rule: $\frac{-1}{5} = 20\%$ Ans. (B less than A)

(4+1)

less than

* When comparing two values.
* When two values are inversly proportional to each other.

EXAMPLE: D = S × T (constant)

S↑ = T↓

S↓ = T↑

Exp. = Price × Consumption (constant)

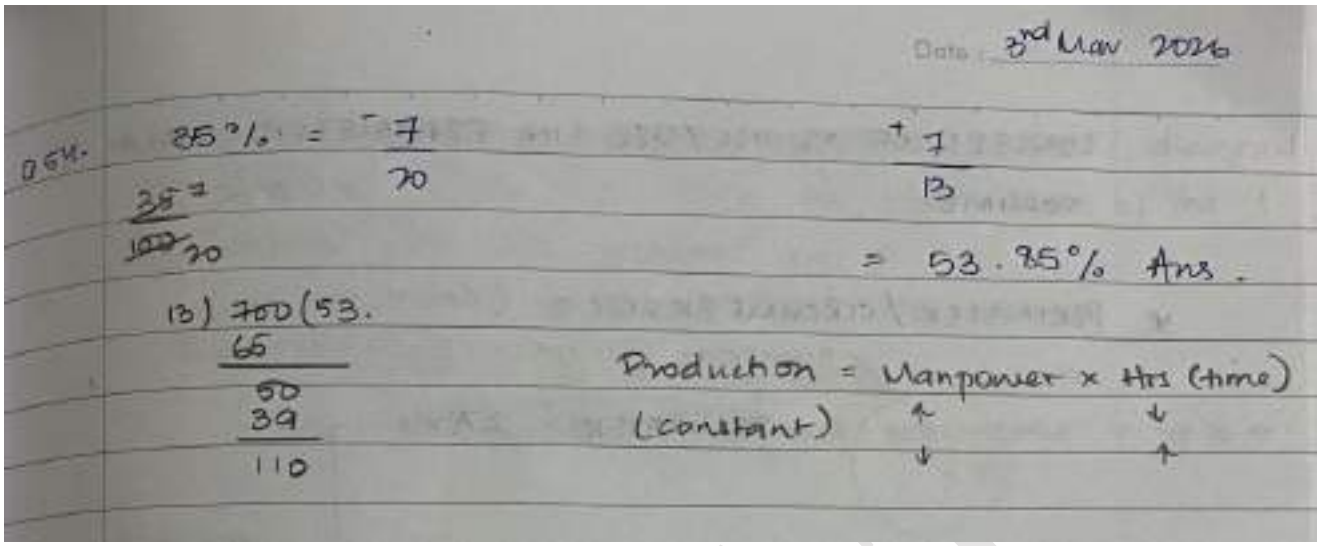
P↑ = C↓

P↓ = C↑

54. Due to fall in manpower, the production in a company decreases by 35%. By what percentage should the working hours be increased to restore the original production?

SSC CPO 11/11/2022 (Shift-01)

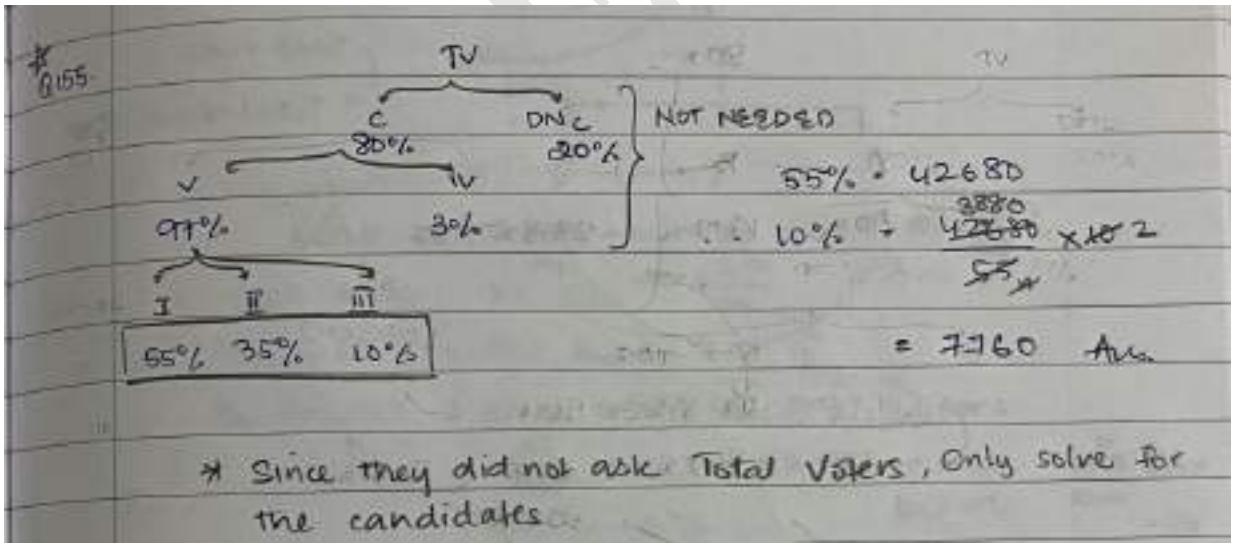
(a) 58.35% (b) 55.83% (c) 53.85% (d) 55.38%



155. In a village election between three candidates, 80% of the voter cast their votes, out of which 3% votes were declared invalid. The winning candidate got 42,680 votes which were 55% of the total valid votes. Find the number of valid votes received by the candidate who stood third, given that he received only 10% of the total valid votes.

SSC CHSL 03/08/2023 (Shift-01)

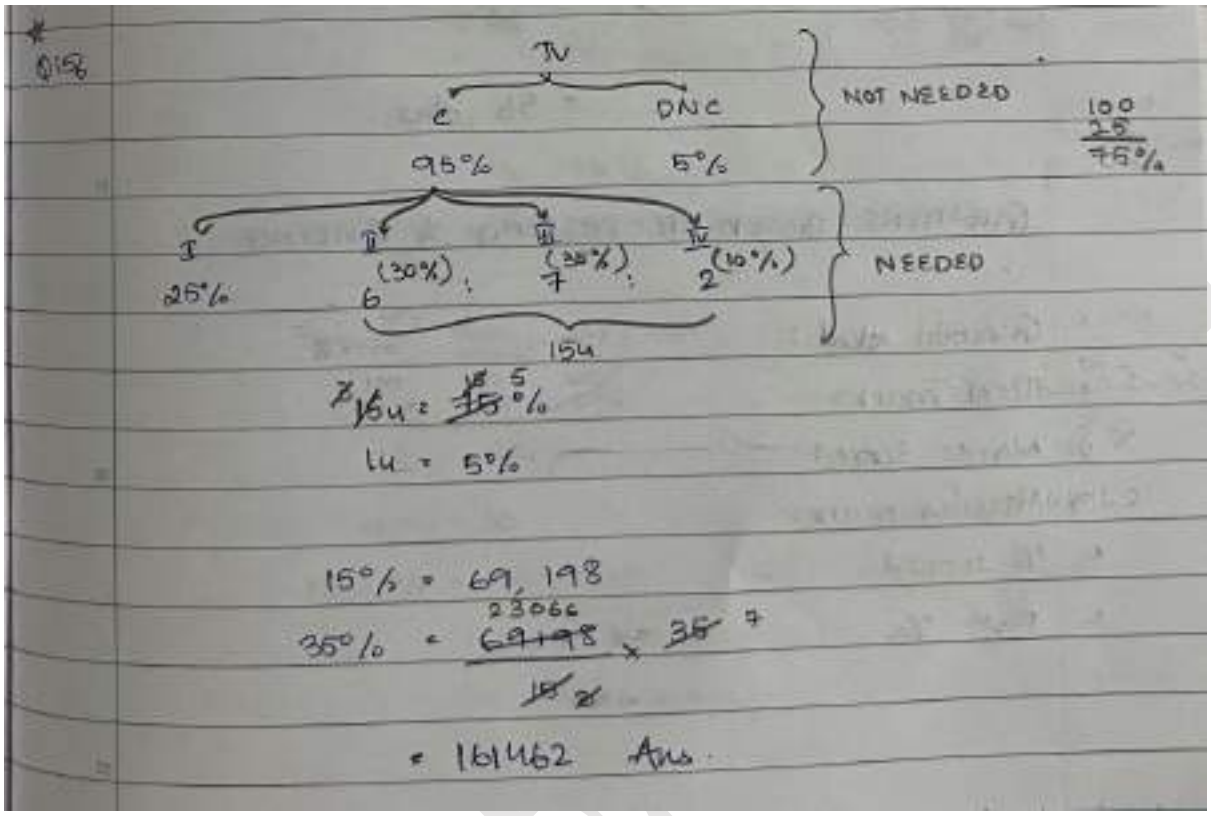
(a) 7760 (b) 7700 (c) 7670 (d) 7680



156. In an election, there were four candidates and 95% of voters cast their votes. One candidate got 25% of the votes while the other three shared the remaining votes in the ratio of 6 : 7 : 2. The difference of votes secured by the candidate standing 3rd to the candidate standing 4th is equal to 69198. Find the votes secured by the winning candidate.

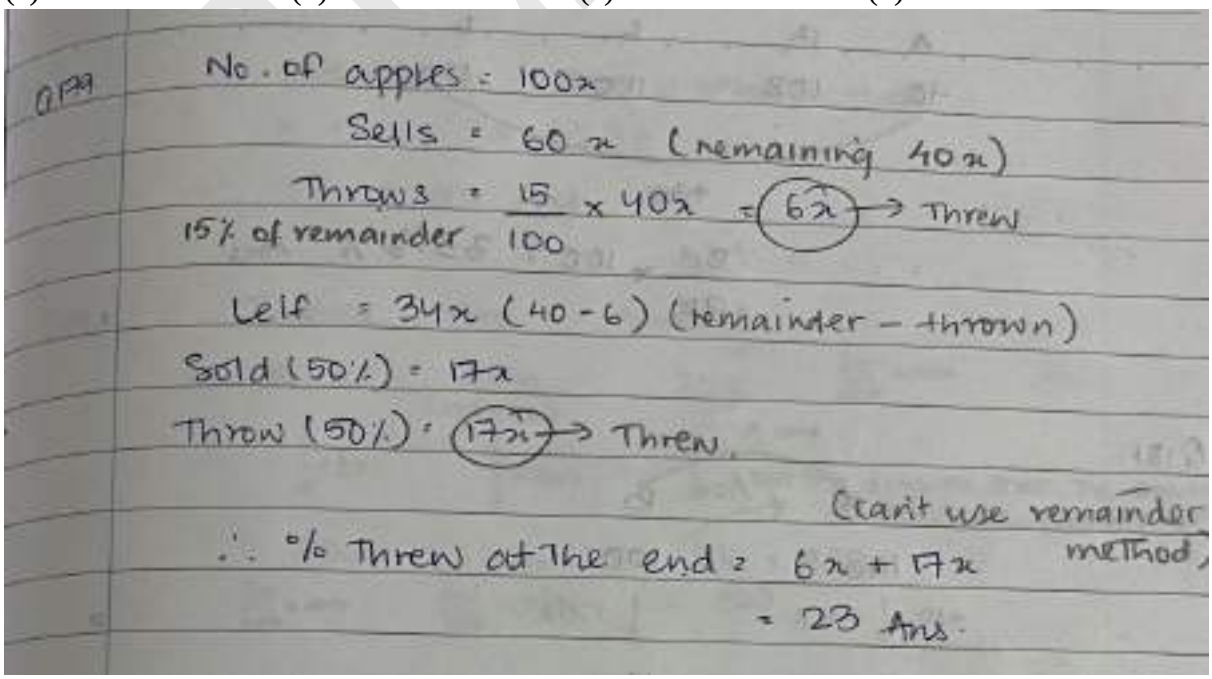
CRPF HCM 27/02/2023 (Shift - 02)

(a) 115330 (b) 138396 (c) 161462 (d) 156458



179. A vendor sells 60% of apples he had and throws away 15% of the remainder. Next day he sells 50% of the remainder and throws away the rest. What % of his apples does the vendor throw?

(a) 17 (b) 21 (c) 23 (d) 29



75. Length of cuboid is increased by 20%, breadth is increased by 30% and height is decreased by 50%. Find effect on volume.

(a) 22% increase (b) 22% decrease (c) 11% increase (d) 11% decrease

INC / DEC IN VOLUME

1D → 1D (No change

1D → 2D Twice successive / Ratio

1D → 3D Thrice successive / Ratio

- cube = $l \times b \times h$ $(side)^3$
- cylinder = $\pi r^2 h$
- cone = $\frac{1}{2} \pi r^2 h$ [unaffected]
- sphere = $\frac{4}{3} \pi r^3$

(Ratio better for these

Q75. $l = 20\% \uparrow +1/5$

$b = 30\% \uparrow +3/10$

$h = 50\% \downarrow -1/2$

	O	:	N
l	5	:	6
b	10	:	13
h	2	:	1
V	100	:	78

-22

$\frac{22}{100} \times 100 = 22\% \downarrow$ Ans.

206. The monthly expenses of a person are $66\frac{2}{3}$% more than her monthly savings. If her monthly income increases by 44% and her monthly expenses increases by 60%, then there is an increase of Rs 1,040 in her monthly savings. What is the initial expenditure (in Rs)?

SSC CGL MAINS 29/01/2022

(a) 9,000 (b) 12,000 (c) 10,000 (d) 13,000

Q206. I = E + S

800x = 500x + 300x

↓+44% ↓+60%

352 = 300x + 92x

44/100 × 800 60/100 × 500

52x = 1040

∴ Initial Exp = 500 × 20

= 10,000 Ans

213. At the beginning of day 1 of a month, Rajesh has 500 eggs. He sells 20% of the eggs by the end of the day and added b% of the eggs at the beginning of the next day and sold 20% of the eggs at the end of the next day. This pattern continued up to the end of the third day of the month when he is left with 1024 eggs. The value of b is equal to:

SSC CHSL 02/07/2024 (Shift-04)

(a) 20 (b) 10 (c) 500 (d) 100

Q213.

500 —20%→ 400

+b%

400×2 800 —20%→ 640

+b%

640×2 1280 —20%→ 1024 eggs.

Sold 3 times / Added 2 time

500 < 1024

∴ b% > 20%

since we added only twice, yet sold 3 times.

when we say something inc. by

100% → + 100/100 = 2/1

500% → + 500/100 = 6/1

Too much.

∴ b% = 100% Ans.

Approximation

APPROXIMATION 02TH Mar 2026

* Simplification → Approximation of values. (Bank Exams)

* BODMAS (application)

Eg: $\sqrt{4} = 2$ $\sqrt{2} = 1$

$\sqrt{121} = 11$ $\sqrt{33} \rightarrow 6$ (closest) $\sqrt{25} = 5$ $\sqrt{36} = 6$ 33 closer to 36

+1 (more than 5)

$3.791\% \rightarrow 4\%$ $(12.96)^2 = 13^2 \rightarrow 169$

$3.21\% \rightarrow 3\%$

$3.5\% \rightarrow \frac{3.5}{100 \times 10} = \frac{7}{200}$

$12.5\% \rightarrow \frac{1}{8}$
(known values)

Q1. $\left(\frac{45}{100} \times 1350\right) + \left(\frac{65}{100} \times 650\right)$

$= \frac{1215}{2} + \frac{845}{2}$

$= \frac{1215 + 845}{2}$

$= \frac{2060}{2}$

$= 1030$ Ans

Q2. $\frac{1}{4} \times x = \sqrt{169} + 5^2$

$\Rightarrow \frac{1}{4} \times x = 13 + 25$

$\Rightarrow x = 38 \times 4$

$\Rightarrow x = 152$ Ans.

Q11. $2222 \times 11 + 3333 \times 11 + 4444 \times 11 + 5555 \times 11 - 6666 \times 11$

$= 11\,(2222 + 3333 + 4444 + 5555 - 6666)$

$= 11 \times 8888$

$= 97768$ Ans

$$\begin{array}{r} 2222 \\ 3333 \\ 4444 \\ 5555 \\ \hline 15554 \\ -\ 6666 \\ \hline 8888 \\ \times\ 11 \\ \hline 97,768 \end{array}$$

Q14. $\frac{\overset{64}{\cancel{768}}}{\cancel{12}} \times 16 - 58$

$= 1024 - 58$

$= 966$ Ans

$$\begin{array}{r} 64 \\ 16 \\ \hline 384 \\ 64 \\ \hline 1024 \\ -\ 58 \\ \hline 966 \end{array}$$

Q13. $x^2 + 114 - 24 \times 5 = 163$

$\Rightarrow x^2 + 114 - 120 = 163$

$\Rightarrow x^2 - 6 = 163$

$\Rightarrow x^2 = 163 + 6$

$\Rightarrow x^2 = 169$

$\Rightarrow x = \sqrt{169}$

$\Rightarrow x = 13$ Ans.

19th Mar 2026

Q10. $470 \times 100 + 338 + 472 - 138 \times 1$

$= 47000 + 338 + 472 - 138$

$= 47810 - 138$

$= 47672$ Ans

Q12. $x\%$ of $\left(\frac{\overset{755}{\cancel{5285}}}{\cancel{7}}\right) = 986 - 533$

$x\%$ of $755 = 453$

$x\% = \frac{453}{755}$

$x = \frac{\overset{3}{\cancel{453}}}{\cancel{755}_{\cancel{100}}} \times \cancel{100}\ 20$

$x = 60$ Ans.

Classification Of Numbers

NUMBER SYSTEM.

Date: 9th Feb 2026

CLASSIFICATION OF NUMBERS

- COMPLEX ×
  - $(a+ib) \rightarrow$ Real + imaginary
  - *REAL (a)
  - IMAGINARY (b)/(ib)

→ Numbers that can be plotted on the no. line (-3 -2 -1 0 1 2 3)

- RATIONAL
  - → The can be expressed in the form of $\frac{p}{q}$, $q \neq 0$
  - INTEGERS
    - -VE INTEGERS
    - WHOLE NOS.
      - ZERO
      - NATURAL NOS
        - ODD
        - EVEN
        - COMPOSITE
        - PRIME
          - CO-PRIME
          - TWIN PRIME
  - FRACTION: 4/5, 6/7
  - DECIMAL: 0.2, 0.5
- IRRATIONAL
  - → Cannot be expressed in the form of $\frac{p}{q}$
  - Eg. $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$

(-3 -2 -1 0 1 2 3)

NEUTRAL

TYPES OF DECIMALS:

RATIONAL
- TERMINATING
- NON-TERMINATING, REPEATING

$1/2 = 0.5$

$1/4 = 0.25$

} TERMINATING DECIMALS > RATIONAL

$1/3 = 0.3333\ldots\infty$ → NON-TERMINATING REPEATING DECIMALS

Eg. $0.1414\ldots\infty$
$0.176176\ldots\infty$

* π

$3.14\ldots\infty$ → IRRATIONAL

* NON-TERMINATING NON-REPEATING DECIMAL

$\frac{22}{7}$

3.14

} RATIONAL

$$\frac{\text{Circumference}}{\text{Diameter}} = \frac{2\pi r}{2r} = \pi$$

* π is an IRRATIONAL no.

Q1 Complex

Q2 Both rational & irrational no

Q3. Irrational

Q4. Sq. root of 7

Date :

Q5. Sq. root of 2.

Q6. $\sqrt{3}+1 \rightarrow (\sqrt{3}+1)^2$

$(a+b)^2 = a^2+b^2+2ab$

$= 3+1+2\sqrt{3}$

$= 4+2\sqrt{3}$ (IRRATIONAL)

$(\sqrt{5})^2 = 5$

$(\sqrt{4.5})^2 = 4.5$

SQUARE ROOT OF DECIMAL NOS.

→ Check if it is a PERFECT SQUARE

(i) if NOT → IRRATIONAL

(ii) if YES → Check:

→ if number of places after decimal is

(i) ODD (1, 3, 5, etc) → IRRATIONAL

(ii) EVEN (2, 4, 6, etc) → RATIONAL

Eg: $\sqrt{0.4} = \sqrt{\frac{4}{10}} = \frac{2}{\sqrt{10}}$ → IRRATIONAL (IR)

$\sqrt{0.04} = \sqrt{\frac{4}{100}} = \frac{2}{10}$ → RATIONAL (R)

10th Feb 2026.

0 → Rational / Irrational ? $0/2, 0/3, 0/4 \rightarrow p/q; q \neq 0$

0.1111... → Non-terminating, repeating (R)

0.1212... → Non-terminating, repeating (R)

0.123123... → Non-terminating, repeating (R)

Unit Digit

Date: 16th Feb 2024

UNIT DIGIT

2 3 1

Hundreds Tens Units / Ones

1254 + 5312 = 6 (units digit)

6789 + 6112 = 7

1113 × 1112 = 6

$2^5 \times 3^7$

NOTE:

5 × odd

5 × 3 = 15

125 × 89 = 5

units digit

5 × even

25 × 4 = 100

125 × 126 = 0

units digit

Q1. 49237 × 3995 × 738 × 83 × 9

7 × 5 × 8 × 3 × 9

40 — 0 × anything else is always 0 in the units digit

= 0 Ans.

Q2. 3 × 38 × 537 × 1256

3 × 8 × 7 × 6

24 42

= 8 Ans.

1.The digit in unit's place of the product

49237 × 3995 × 738 × 83 × 9 is :

(a) 0 (b) 7 (c) 6 (d) 8

2. The unit digit in 3 × 38 × 537 × 1256 is :

3 × 38 × 537 × 1256

(a) 4 (b) 2 (c) 6 (d) 8

3. What is the unit digit in the multiplication of

$1 \times 3 \times 5 \times 7 \times 9 \ldots \times 999$?

UPSC CSE (CSAT) 25/05/2025

(a) 1 (b) 3 (c) 5 (d) 9

4. Find the unit digit of the product of all the prime number.

(a) 0 (b) 1 (c) 2 (d) 5

5. For any natural number n, what is always the last digit of the result $6^n - 5^n$?

NTPC CBT-1, 28/12/2020 (Shift-02)

(a) 1 (b) 3 (c) 7 (d) 5

Date :

Q 3. $1 \times 3 \times 5 \times 7 \times 9 \ldots \times 999$. UPSC CSE (CSAT) 25/05/2020

$= 5$ Ans.

Q 4. Units digit of all Prime Nos (2) 3, (5) 7 ...

$= 0$ Ans. 10

Q5 natural no = n NTPC CBT - 1, 28/12/2020 (Shift-

units digit $= 6^n - 5^n$ $6^1 - 5^1 = 1$

$= 1$ Ans $6^2 - 5^2 = 1$

CONCEPT OF CYCLICITY: After a certain frequency or cycle length.

(Power based Questions)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

$\Rightarrow 0^1 = 0$, $0^2 = 0$, $0^3 = 0$ } Cyclicity of 0 = 1

$\Rightarrow 1^1 = 1$, $1^2 = 1$, $1^3 = 1$ } Cyclicity of 1 = 1

$\Rightarrow 5^1 = 5$, $5^2 = 25$, $5^3 = 125$ } Cyclicity of 5 = 1

5^4: $5^2 \times 5^2$, 25×25